

Dipartimento di Ingegneria Informatica, Modellistica, Elettronica e Sistemistica

Corso di Laurea Magistrale in Ingegneria Elettronica

Design e caratterizzazione di IP-Core su FPGA per funzioni di attivazione in modelli di Intelligenza Artificiale

Relatrice:

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Candidato:

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Matr. 247705

ANNO ACCADEMICO 2025/2026

Introduzione

CONTESTO:

Transformer
Computer Vision



PROBLEMI:

Calcolo computazionalmente oneroso;

L'implementazione efficiente su FPGA rappresenta una sfida progettuale.

SOLUZIONE:

Acceleratore hardware

FUNZIONI DI ATTIVAZIONE

VALIDAZIONE

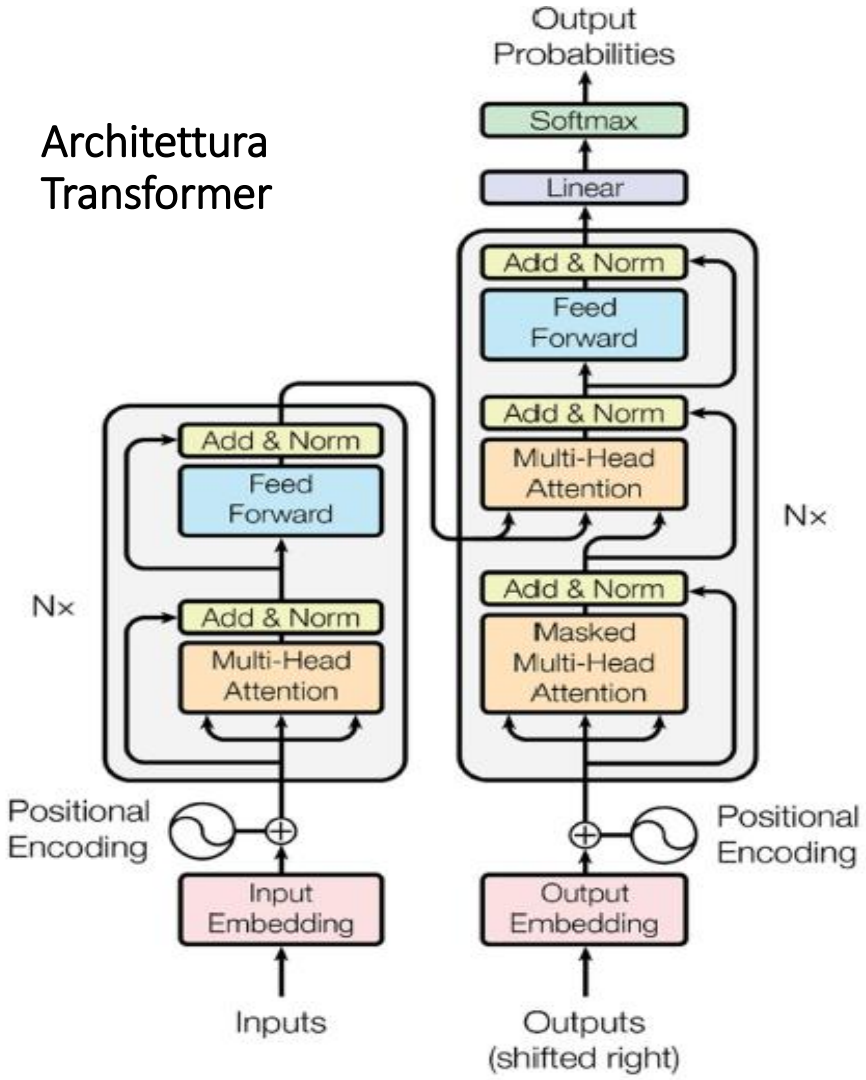
Simulazioni post-layout;
Confronto hardware/software

SVILUPPO ARCHITETTURA E OTTIMIZZAZIONI

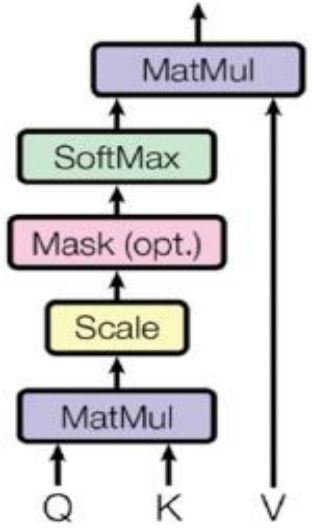
CARATTERIZZAZIONE

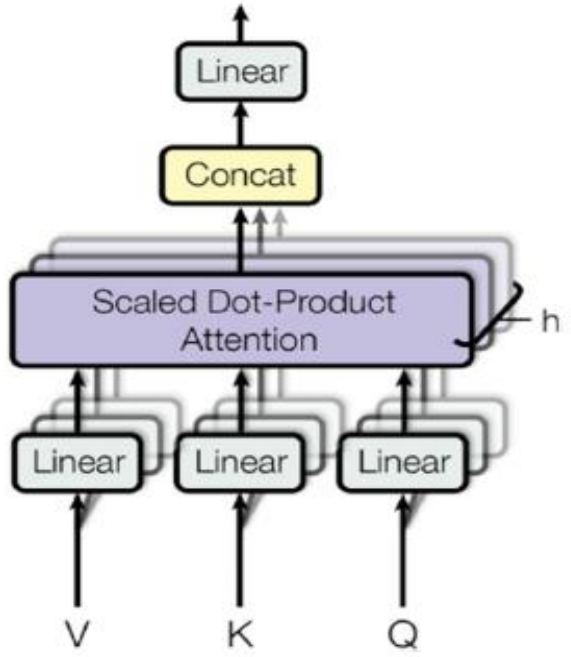
Accuratezza numerica;
Impiego risorse;
Timing;
Consumo energetico;
Scalabilità

Architettura Transformer



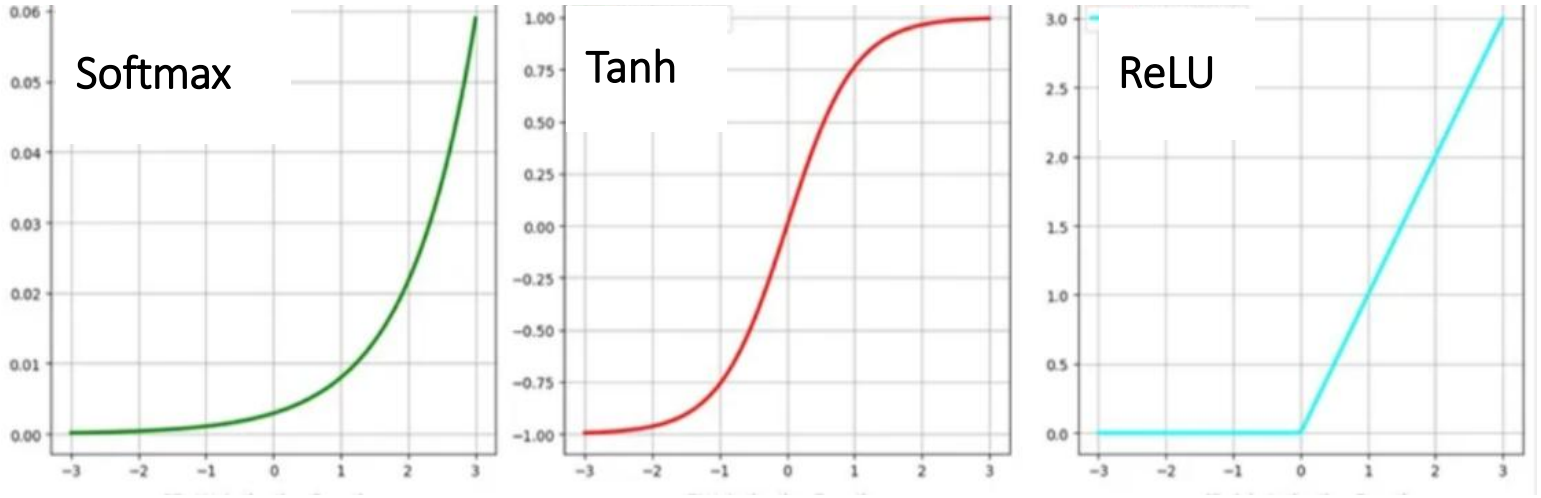
Meccanismo di Self-Attention

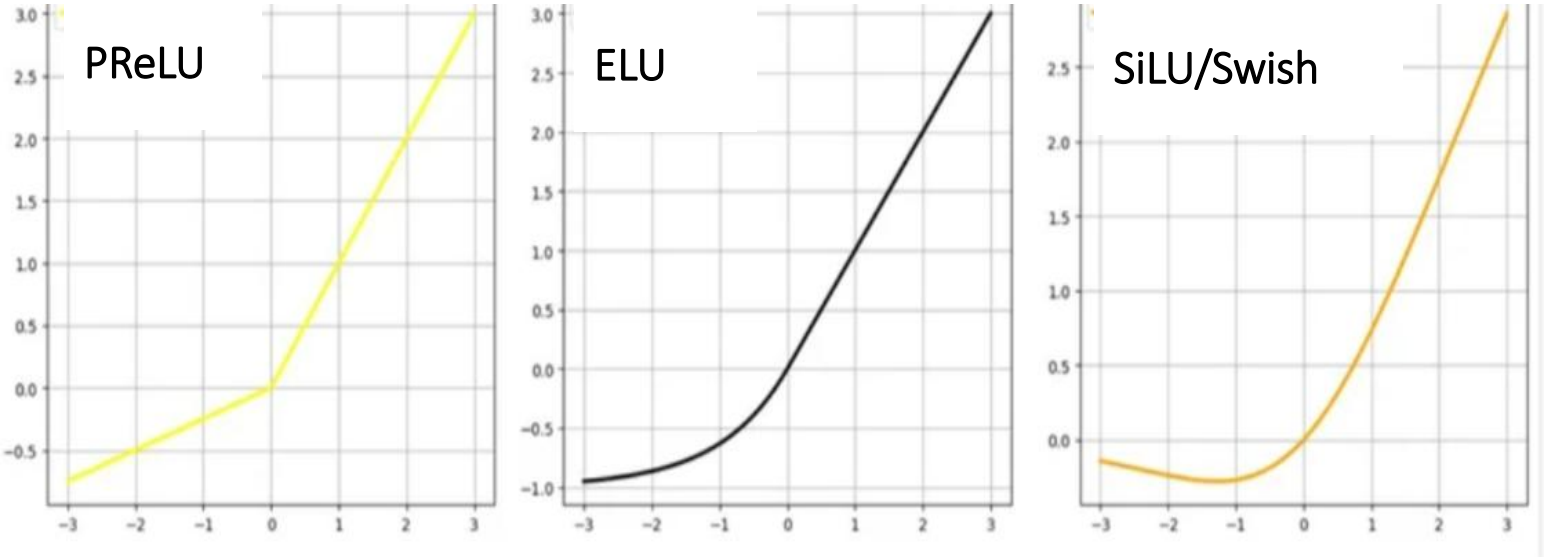


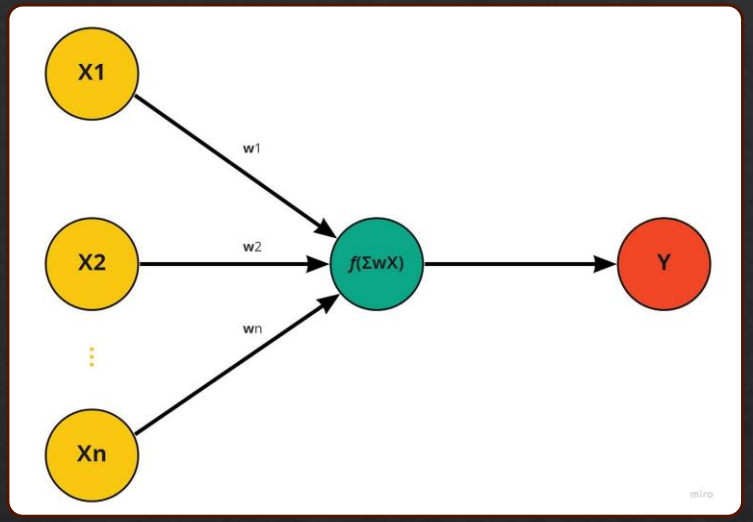


$$Attention(Q, K, V) = softmax\left(\frac{Q\,K^T}{\sqrt{d_k}}\right)V$$

Funzioni di Attivazione





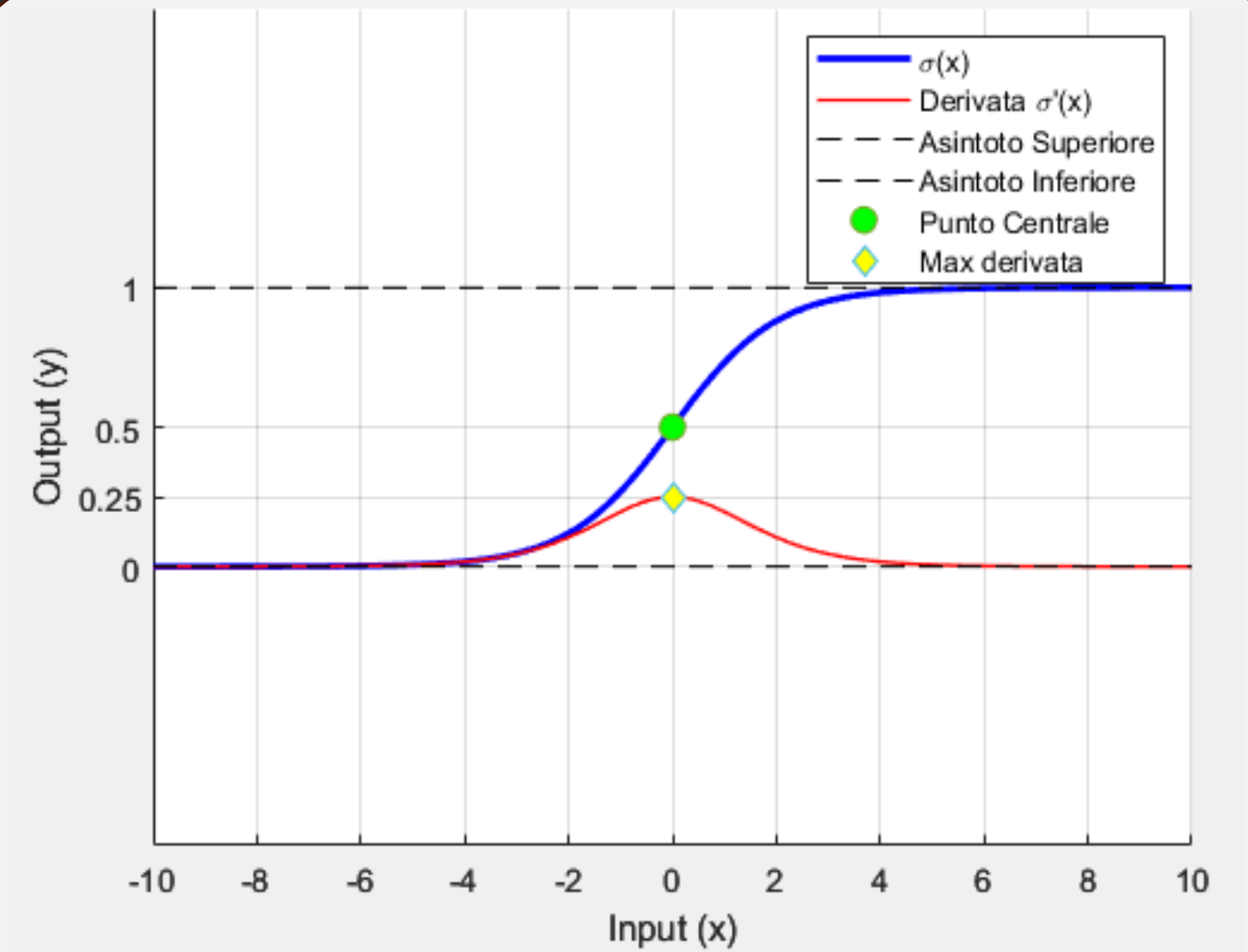


Introducono **non-linearità**, permettendo alla rete di decidere cosa è rilevante

(es. attivare un neurone o spegnerlo).

Senza funzioni di attivazione, una rete neurale sarebbe incapace di apprendere modelli complessi come il linguaggio naturale o le relazioni temporali.

Caso studio: Funzione Sigmoide



$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Compito:

mappare l'intero dominio dei numeri reali $[-\infty, +\infty]$ nell'intervallo $(0,1)$.

Principali utilizzi:

- **Gating:** Se il valore in input è molto alto, il gate si apre (valore 1); se è molto basso, si chiude (valore 0). Questo permette alla rete neurale di decidere dinamicamente quanta informazione far passare attraverso i layer.
- **Interpretazione Probabilistica:** Poiché l'output è compreso tra 0 e 1, la sigmoide viene spesso usata per rappresentare probabilità

Approccio utilizzato

CORDIC

Approccio di tipo iterativo

PRO:
- Eccellente accuratezza;
- Minima occupazione di area;
- Non richiede moltiplicatori hardware (DSP)

CONTRO:
- Elevata latenza.

LUT-BASED

Valori pre-calcolati

PRO:
- Accuratezza ideale;
- Latenza minima e deterministica;
- Zero logica aritmetica complessa (nessun moltiplicatore)

CONTRO:
- Se non ottimizzato, la dimensione della memoria aumenta in modo esponenziale con il numero di bit.

Piecewise Linear (PWL)

Approssimazione mediante rette

PRO:
- Buona accuratezza;
- Minima occupazione di area;
- Latenza fissa

CONTRO:
- Logica di decodifica complessa;
- Richiede moltiplicatori hardware (DSP);
- Errore geometrico intrinseco in giunzione segmenti.

Flusso di lavoro

PYTHON

- Scripting;
- Automazione;
- Golden Model.

AMD/Xilinx VIVADO

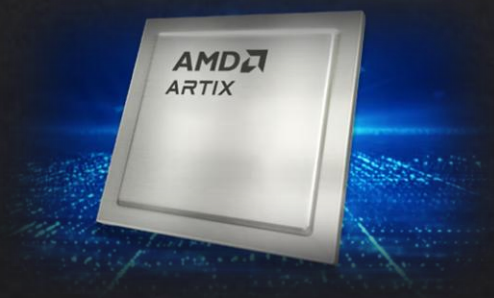
- Descrizione Hardware;
- Sintesi e Mappatura Hardware;
- Simulazione e validazione;
- Caratterizzazione.

MATLAB

- Plotting grafici;
- Test eseguiti su CPU
- Controllo delle metriche di errore (MAE, errore massimo).

Microsoft Excel

Riferimento iniziale



Xilinx Artix-7 XC7A200TSBV484-1

CIFAR II

Dataset reale

Architetture

Baseline (V1)

Memoria: RAM distribuita

Protezione overflow: Assente ✗

- Input: Q5.11
- Output: Q5.11

256 campioni di $\sigma(x)$ con x ∈ $[-4; 4]$

- $\Delta(x) = 0.03125$
- Risoluzione: 256 punti

x4

Intermedia (V2)

Memoria: BRAM dedicata

Protezione overflow: Logica di clipping a monte ✓

- Input: Q5.11
- Output: Q5.11

1024 campioni di $\sigma(x)$ con x ∈ $[-4; 4]$

- $\Delta(x) = 0.0078125$
- Risoluzione: 1024 punti

x2

Avanzata (V3)

Memoria: BRAM dedicata

Protezione overflow: Logica di clipping a monte ✓

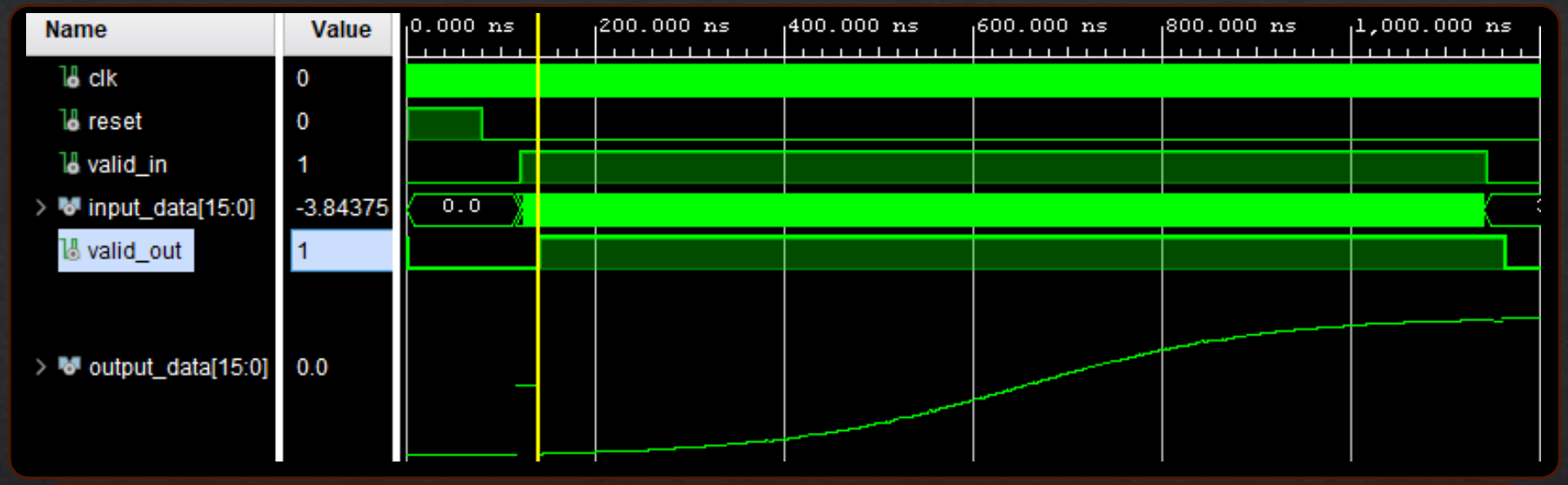
- Input: Q5.11
- Output: Q1.15

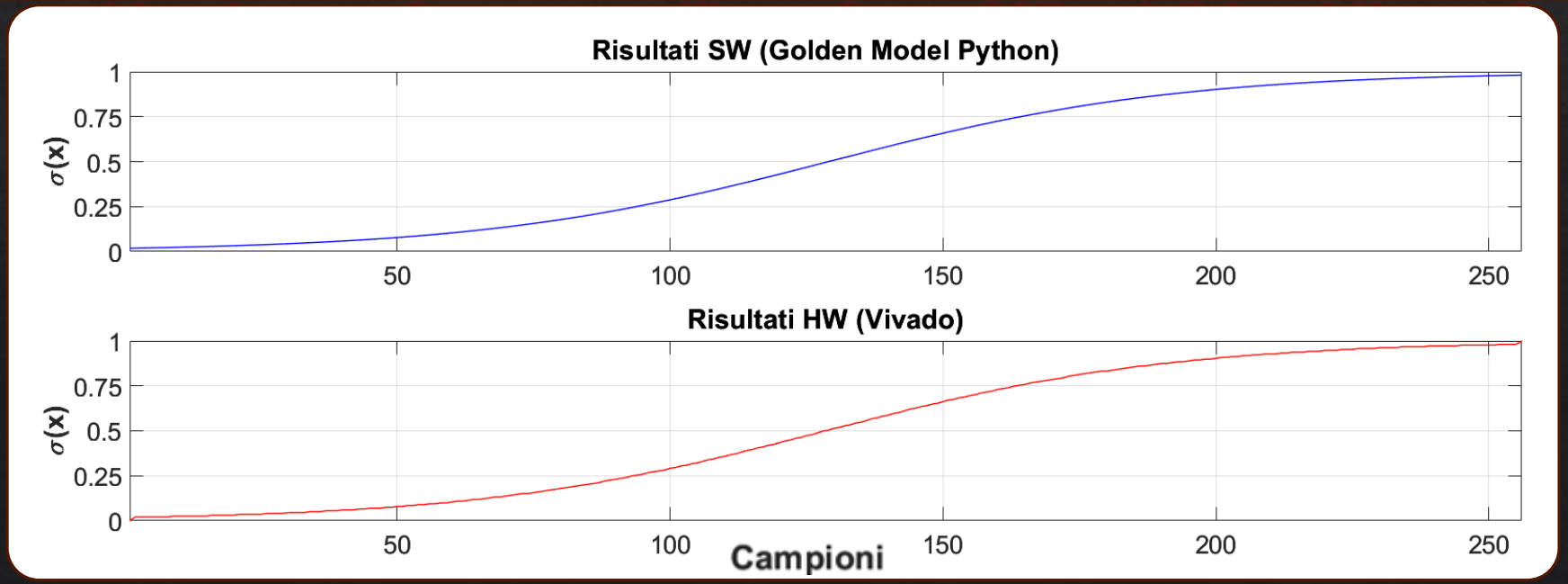
1024 campioni di $\sigma(x)$ con x ∈ $[0; 8]$ e sfruttamento simmetria sigmoide $\sigma(-x) = 1 - \sigma(x)$ per ricostruire il dominio negativo

- $\Delta(x) = 0.0078125$
- Risoluzione: 2048 punti

Test e validazione

- Generazione testbench
- Simulazione timing post-layout
- Generazione file di testo
- Creazione Golden Model
- Confronto accuratezza algoritmica





Confronto hardware vs software

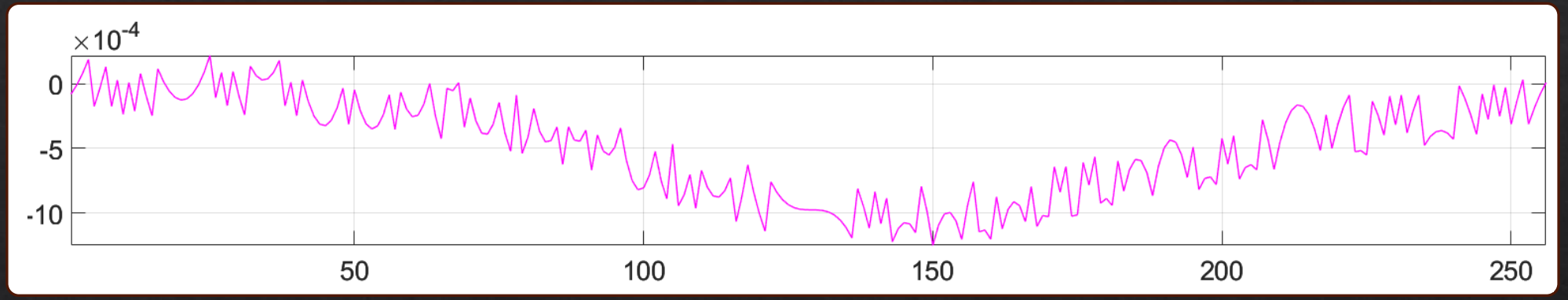
V1

MAE: 0.0023995273
Errore Massimo:
0.0185465645



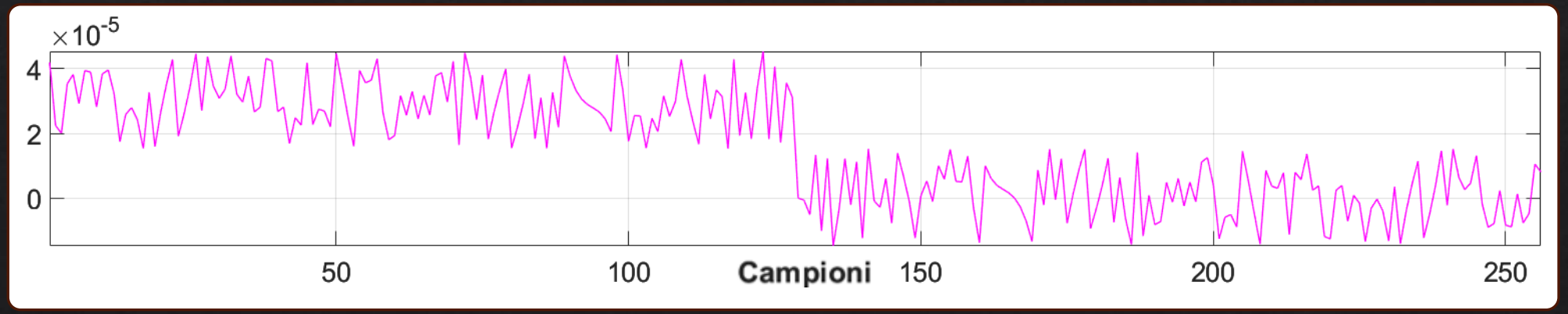
V2

MAE: 0.0005023378
Errore Massimo:
0.0012504682



V3

MAE: 0.0000185288
Errore Massimo:
0.0000452953



Report di utilizzo risorse

V1

Slice LUTs (134600)	Slice Registers (269200)	F7 Muxes (67300)	F8 Muxes (33650)	Slice (33650)	LUT as Logic (134600)	Bonded IOB (285)	BUFGCTRL (32)
43	53	16	8	21	43	28	1
0.03%	0.02%	0.02%	0.02%	0.06%	0.03%	9.82%	3.13%

V2

Slice LUTs (134600)	Slice Registers (269200)	Slice (33650)	LUT as Logic (134600)	Block RAM Tile (365)	Bonded IOB (285)	BUFGCTRL (32)
16	65	22	16	0.5	36	1
0.01%	0.02%	0.07%	0.01%	0.14%	12.63%	3.13%

V3

Slice LUTs (134600)	Slice Registers (269200)	Slice (33650)	LUT as Logic (134600)	Block RAM Tile (365)	Bonded IOB (285)	BUFGCTRL (32)
39	85	26	39	0.5	36	1
0.03%	0.03%	0.08%	0.03%	0.14%	12.63%	3.13%

Report di timing

*Frequenza di riferimento: 250 MHz

	Massima velocità teorica	Massima velocità validata
V1	383 MHz	307 MHz
V2	309 MHz	285 MHz
V3	282 MHz	260 MHz

V1			
Worst Negative Slack (WNS):	1,394 ns	Worst Hold Slack (WHS):	0,157 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	44	Total Number of Endpoints:	44

V2			
Worst Negative Slack (WNS):	0,773 ns	Worst Hold Slack (WHS):	0,166 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	68	Total Number of Endpoints:	68

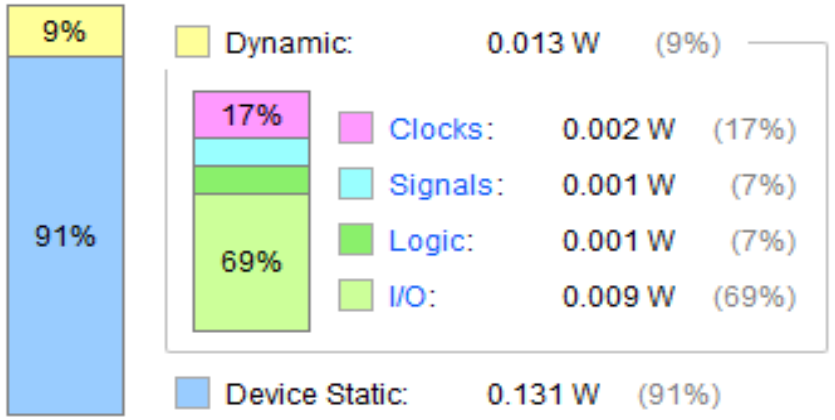
V3			
Worst Negative Slack (WNS):	0,453 ns	Worst Hold Slack (WHS):	0,108 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	88	Total Number of Endpoints:	88

Report di potenza

*Frequenza di riferimento: 250 MHz

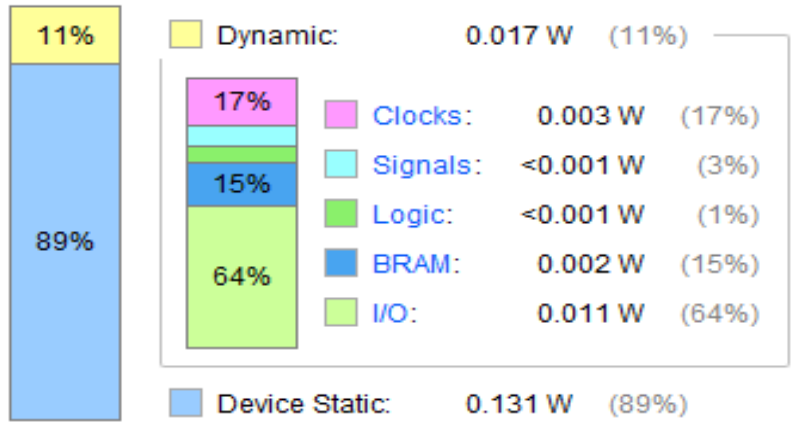
V1

Total On-Chip Power:	**0.144 W**
Design Power Budget:	**Not Specified**
Process:	typical
Power Budget Margin:	**N/A**
Junction Temperature:	**25,4°C**
Thermal Margin:	59,6°C (22,7 W)
Ambient Temperature:	25.0 °C
Effective ϑJA:	2,6°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	High



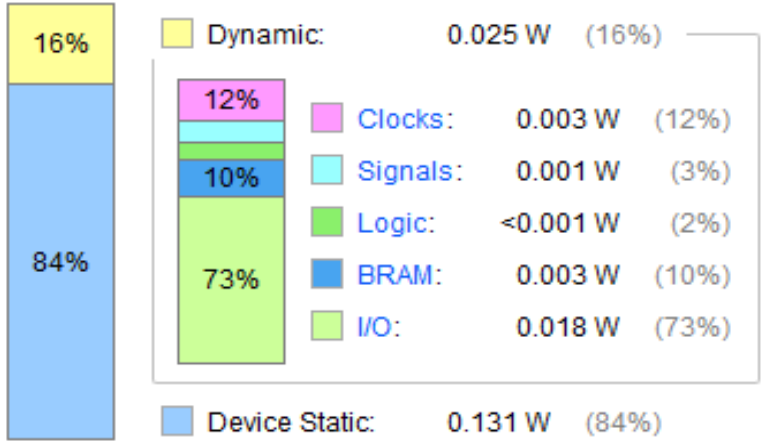
V2

Total On-Chip Power:	**0.148 W**
Design Power Budget:	**Not Specified**
Process:	typical
Power Budget Margin:	**N/A**
Junction Temperature:	**25,4°C**
Thermal Margin:	59,6°C (22,7 W)
Ambient Temperature:	25.0 °C
Effective ϑJA:	2,6°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	High



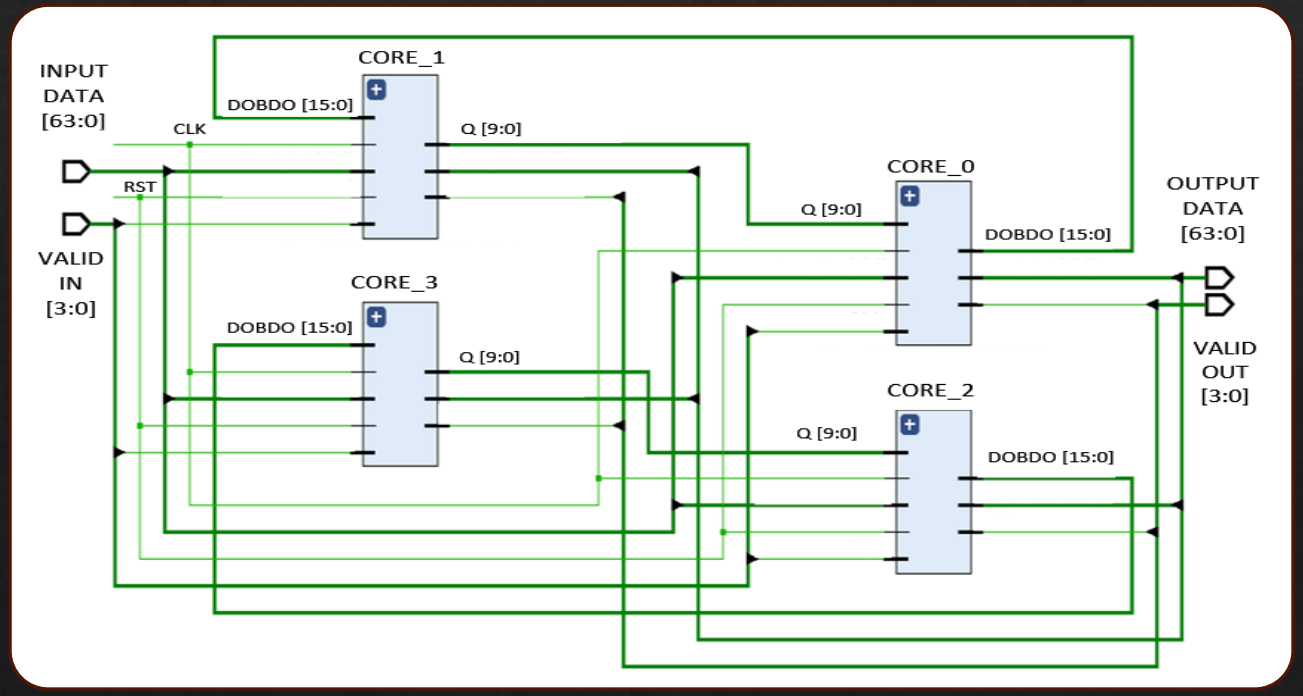
V3

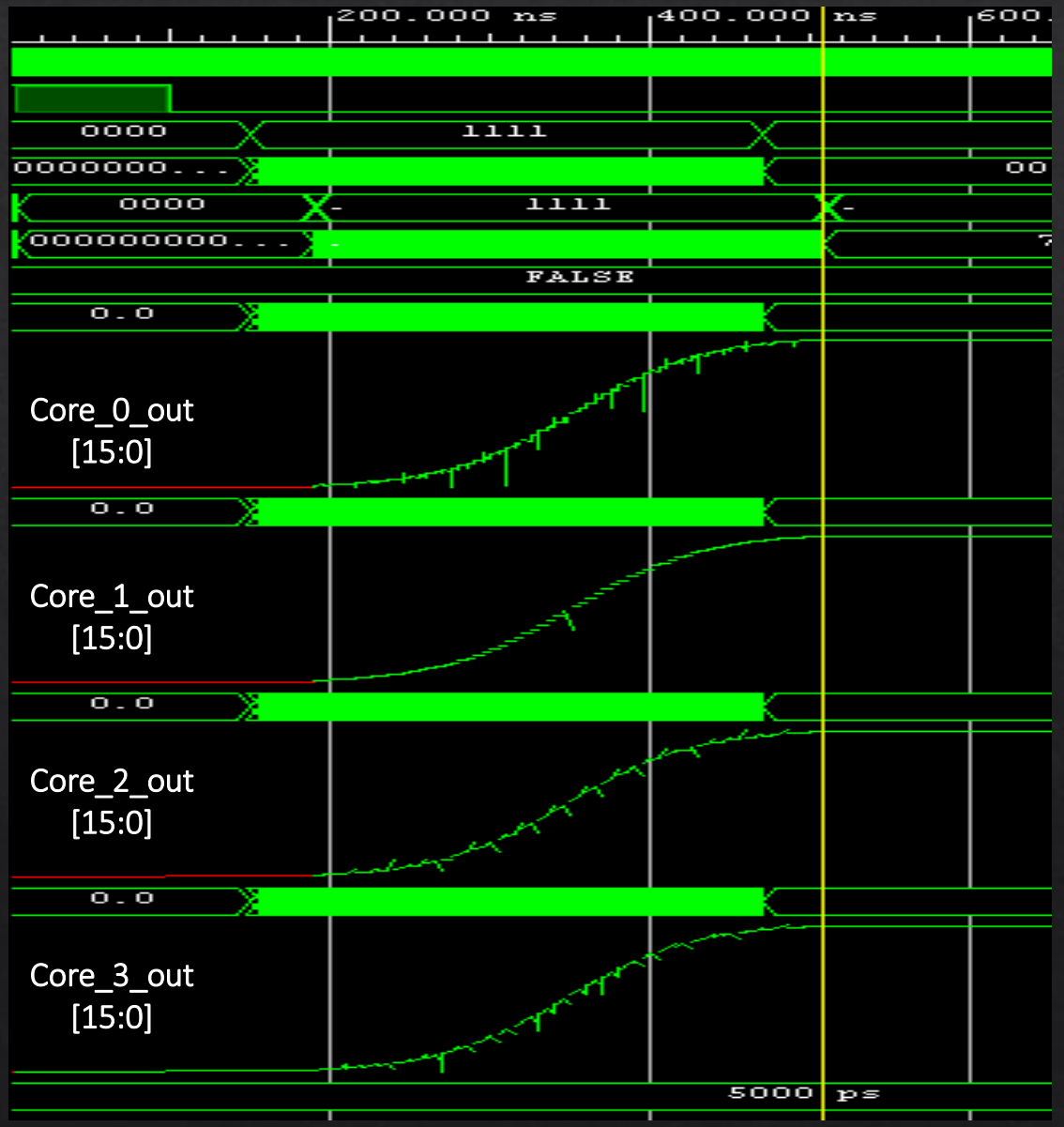
Total On-Chip Power:	**0.156 W**
Design Power Budget:	**Not Specified**
Process:	typical
Power Budget Margin:	**N/A**
Junction Temperature:	**25,4°C**
Thermal Margin:	59,6°C (22,7 W)
Ambient Temperature:	25.0 °C
Effective ϑJA:	2,6°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	High



Parallelismo


```
entity Sigmoid_MultiCore_Top is
    generic (
        NUM_CORES : integer := 4 --SELEZIONARE QUI NUMERO DI CORES
    );
    Port (
        clk, reset : in  STD_LOGIC;
        valid_in : in  STD_LOGIC_VECTOR(NUM_CORES-1 downto 0);
        input_data : in  STD_LOGIC_VECTOR((NUM_CORES * 16) - 1 downto 0);
        valid_out : out STD_LOGIC_VECTOR(NUM_CORES-1 downto 0);
        output_data : out STD_LOGIC_VECTOR((NUM_CORES * 16) - 1 downto 0)
    );
end Sigmoid_MultiCore_Top;
```





Scalabilità (Analisi OOC)

Moduli paralleli	1	2	4	8	16	32	64	128	256	512	1024
FF	85	170	340	680	1360	2720	5440	10880	21760	43520	93508
LUT	31	61	122	244	488	976	1953	3907	7813	15629	79772
BRAM 36 Kb	0.5	0.5	1	2	4	8	16	32	65	130	185,5
Frequenza massima teorica (MHz)	320	300	266	269	267	280	260	266	262	259	132
Throughput massimo teorico (GOP/s)	0.320	0.600	1.064	2.152	4.272	8.960	16.640	34.048	67.072	132.608	135.168
Potenza Totale (W)	0.127	0.133	0.144	0.166	0.190	0.238	0.333	0.508	1.164	2.144	3,173
Potenza dinamica/ statica (%)	4/96	8/92	15/85	27/73	35/65	48/52	63/37	75/25	89/11	93/7	95/5

Saturazione

FPGA vs CPU

Elaborazione di 216.000 campioni provenienti da tensori dataset CIFAR-2

Architettura di Calcolo	CPU S-T (Loop For)	CPU M-T (Vettorizzato)	IP Core FPGA 1 PE (OOC)	IP Core FPGA 2 PE (OOC)	IP Core FPGA 4 PE (OOC)	IP Core FPGA 8 PE (OOC)	IP Core FPGA 16 PE (OOC)	IP Core FPGA 32 PE (OOC)	IP Core FPGA 64 PE (OOC)
Tempo di Elaborazione (ms)	≈ 2.96	≈ 0.74	1.08	0.54	0.27	0.135	0.0675	0.0338	0.0169
Throughput (MSamples/s)	≈ 72.97	≈ 308.57	200	400	800	1600	3200	6400	12800
Consumo Massimo (W)	≈ 35.82	≈ 50.44	0.128	0.133	0.144	0.166	0.190	0.238	0.333

CPU di test: AMD Ryzen 5 7535H

Conclusioni

Target:

- Offloading eterogeneo
- Sistemi autonomi
- Edge AI real-time
- Reti YOLO
- IOT
- ViT

IP Core FPGA LUT-Based

- Elevata accuratezza
- Minimo utilizzo di risorse
- Latenza costante
- Elevato throughput
- Elevata efficienza energetica
- Scalabilità

Sviluppi futuri:

- Scaling fino a 64 moduli paralleli per configurazioni UltraScale+ (AXI 1024-bit)
- Ottimizzazione precisione Q4.4 / Q1.7 per ulteriore densità logica

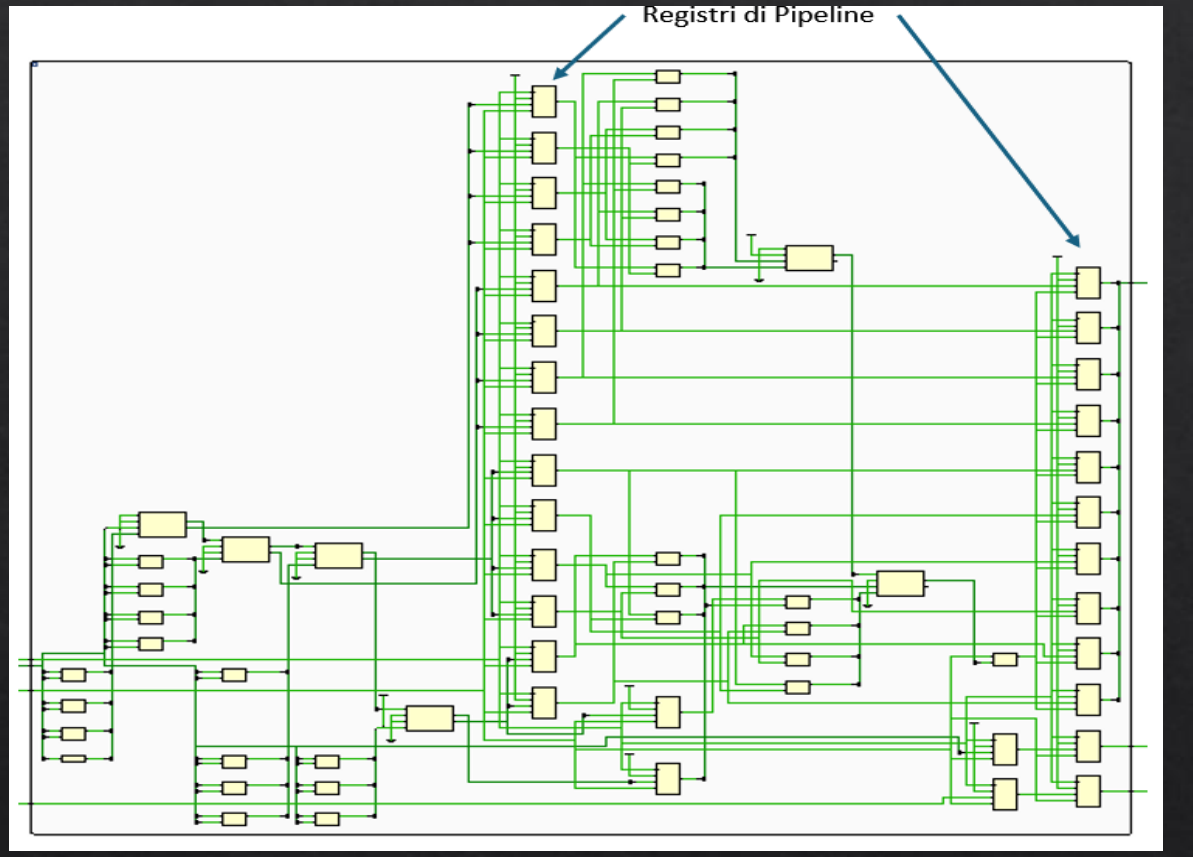
UNIVERSITÀ DELLA CALABRIA

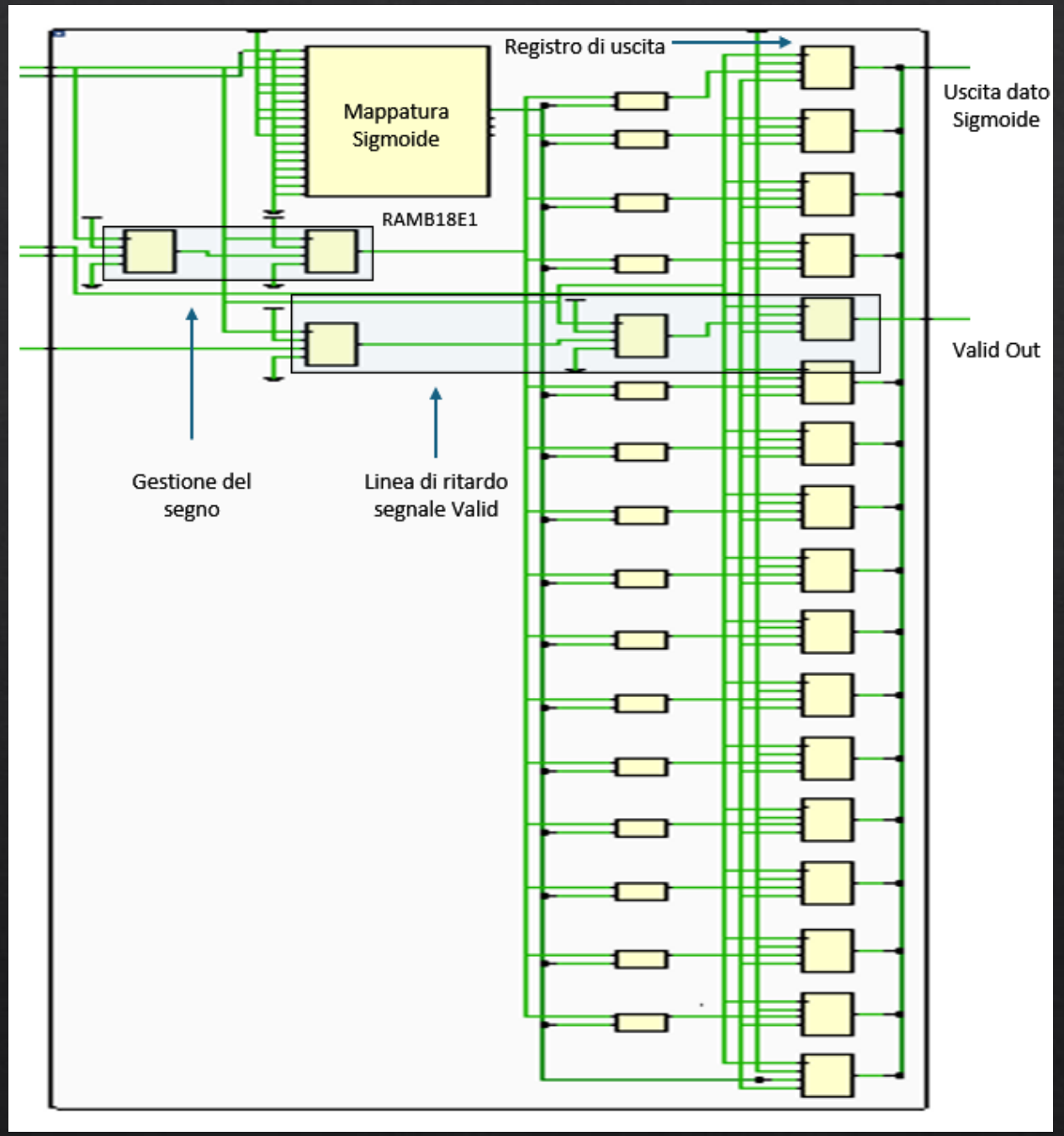
DIPARTIMENTO DI INGEGNERIA INFORMATICA, MODELLISTICA, ELETTRONICA E SISTEMISTICA

Hidden Slides

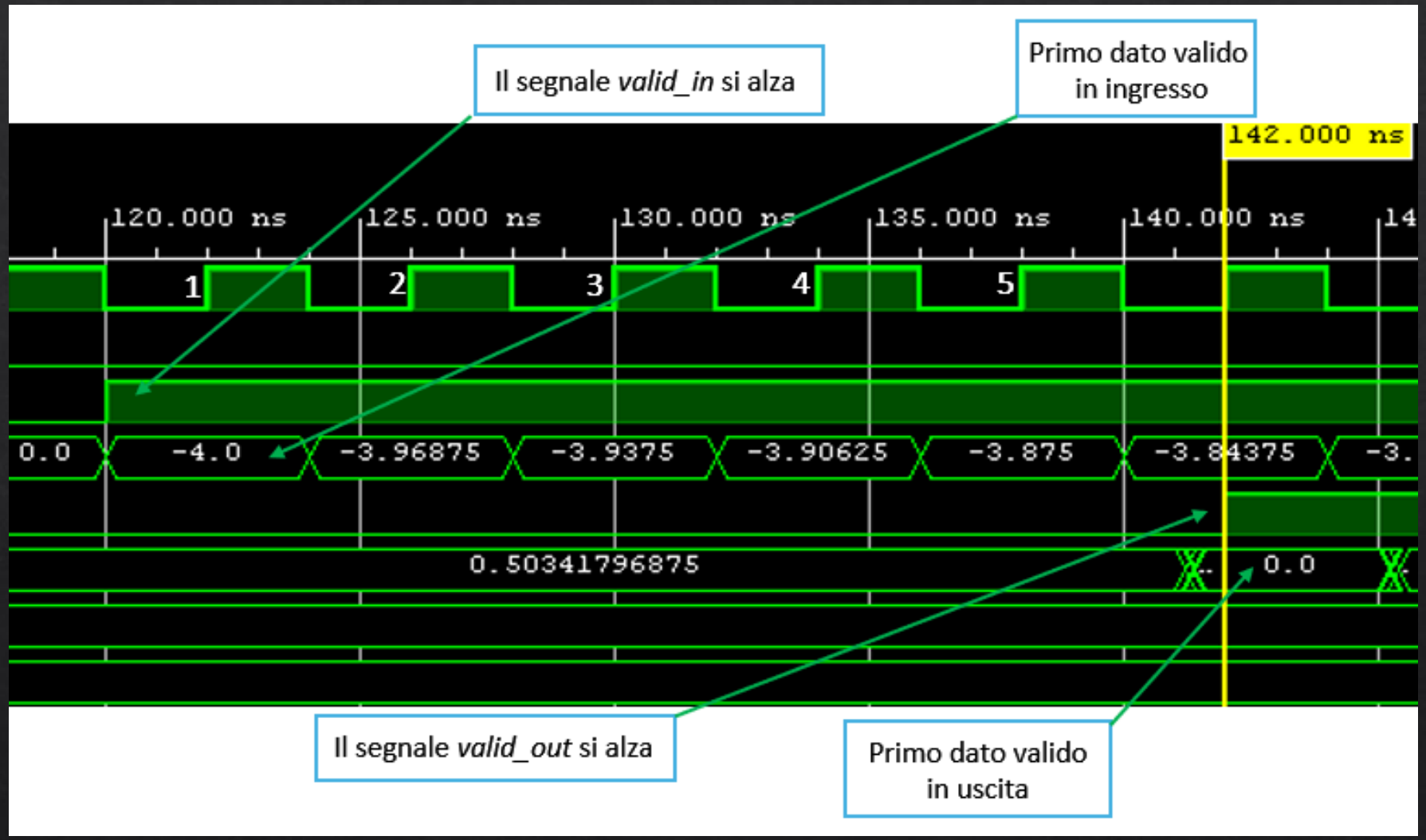


V3

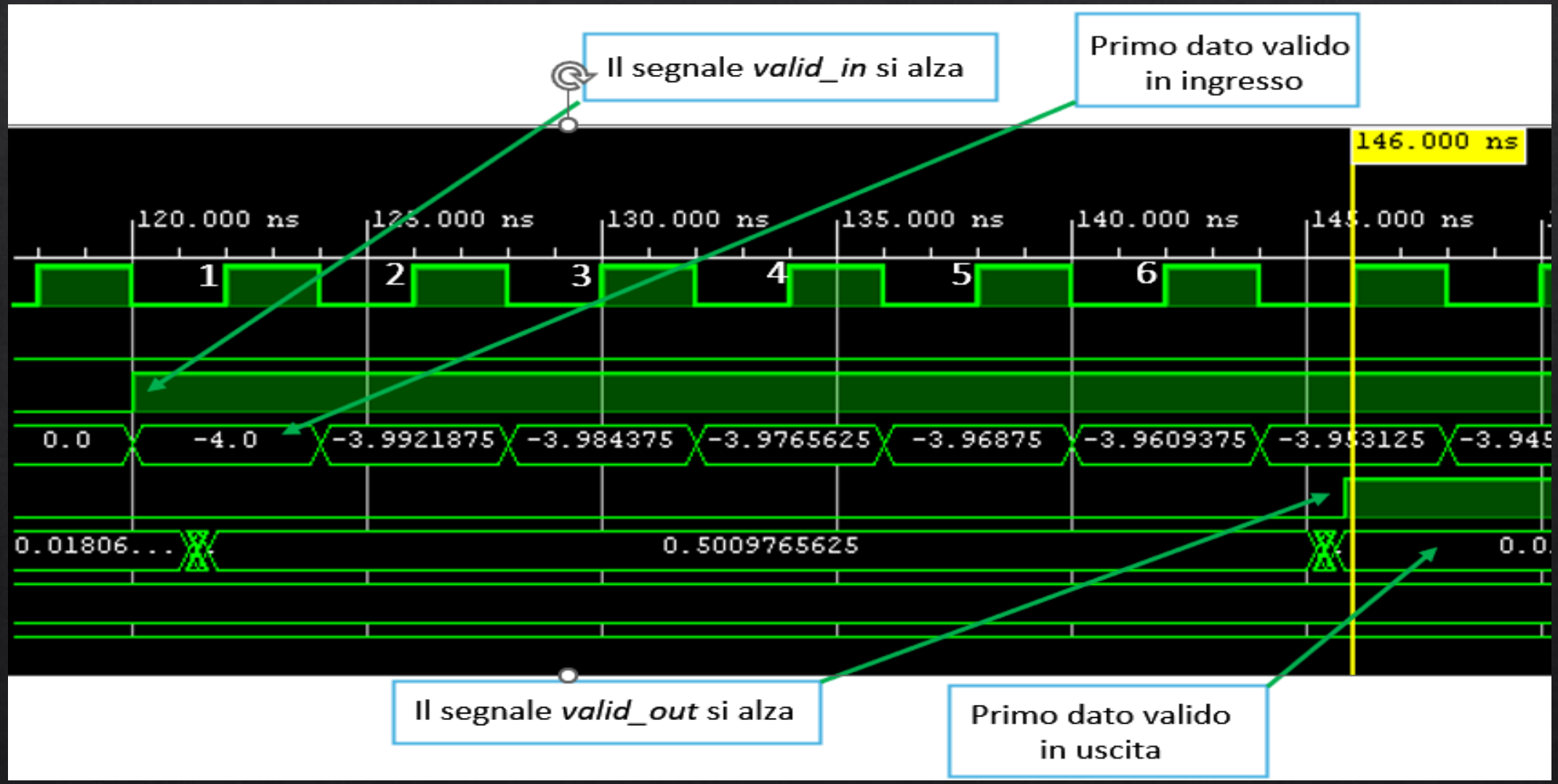

Registri di Pipeline


Registro di uscita
Mappatura Sigmoide
RAMB18E1
Uscita dato Sigmoide
Valid Out
Gestione del segno
Linea di ritardo segnale Valid

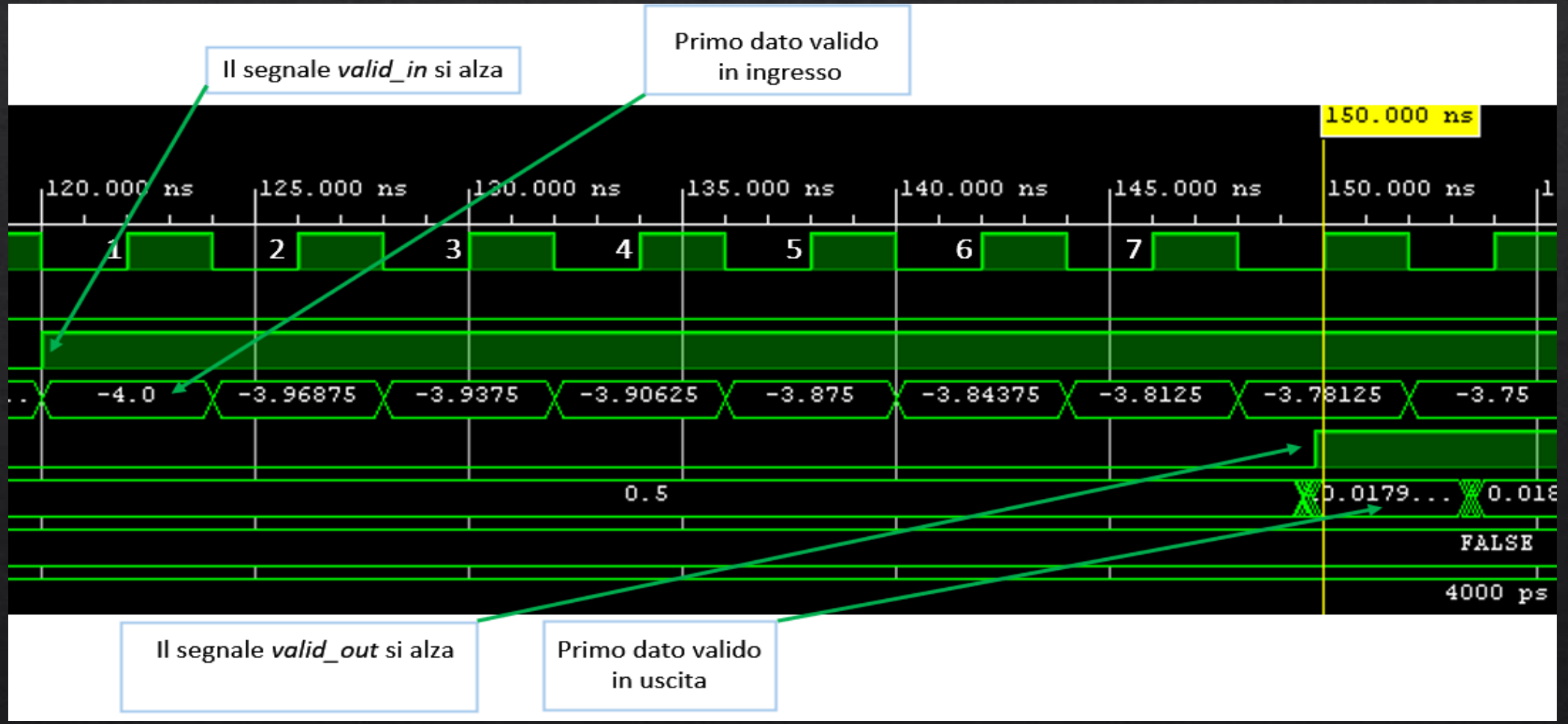
Pipeline a 5 stadi



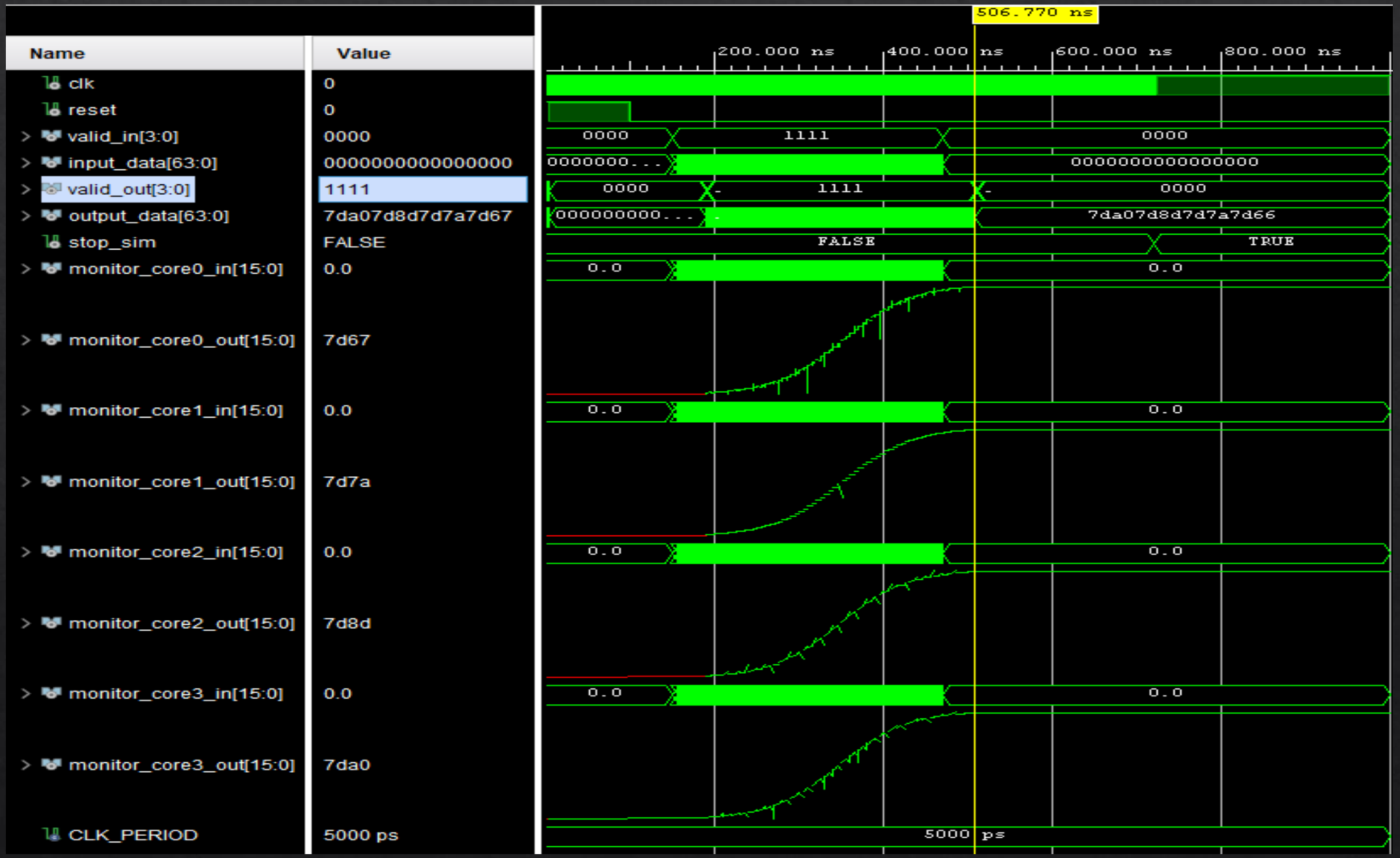
Pipeline a 6 stadi



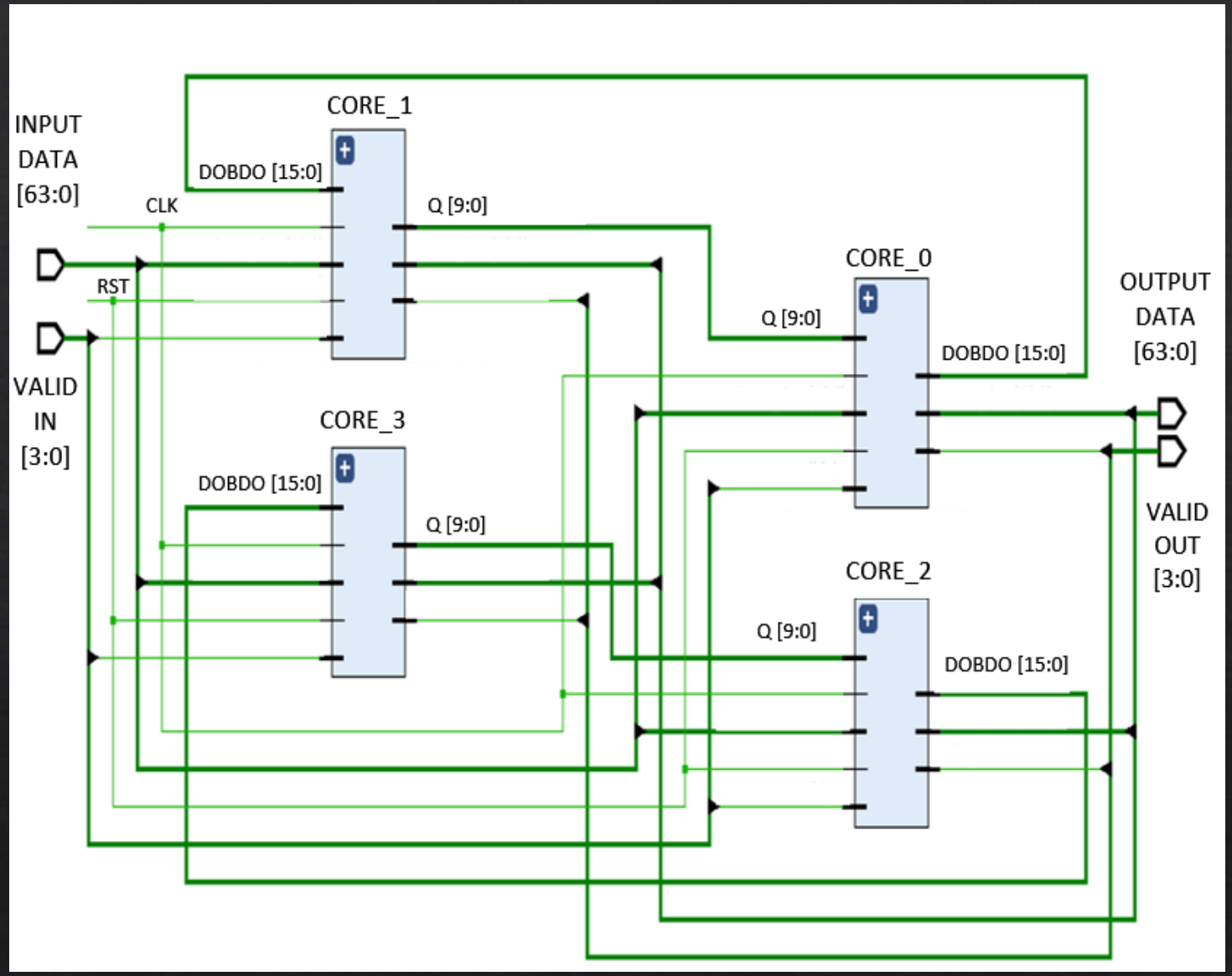
Pipeline a 7 stadi



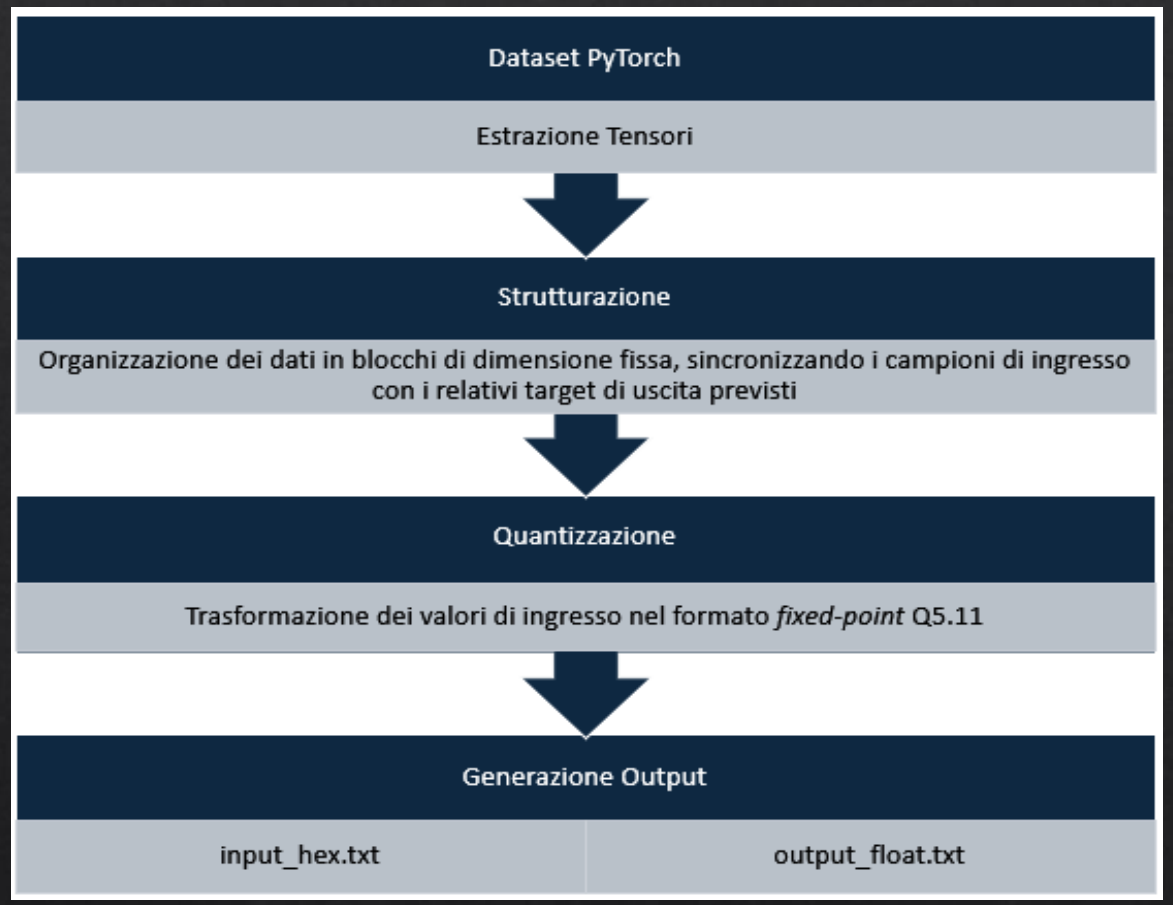
Dettaglio Parallelismo

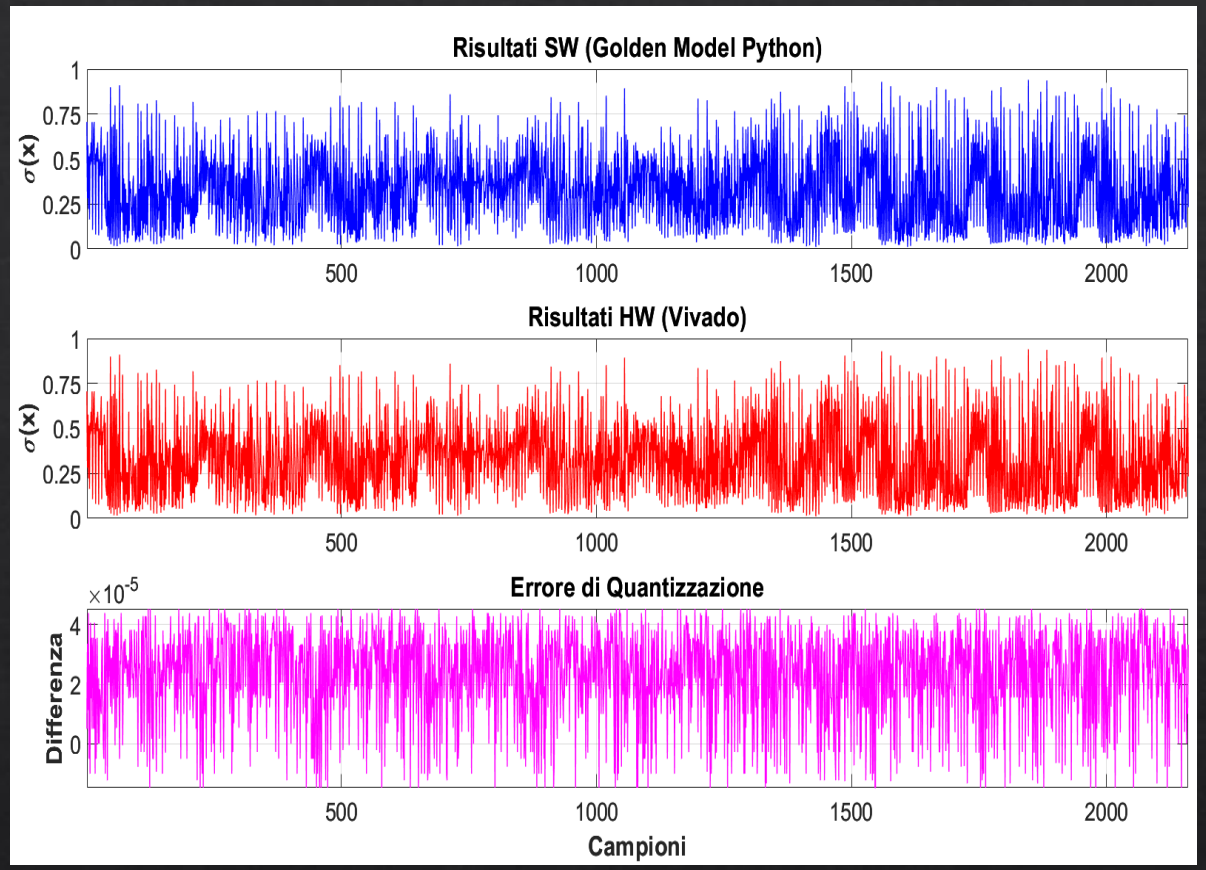


Dettaglio Parallelismo



Dataset tensori CIFAR-2

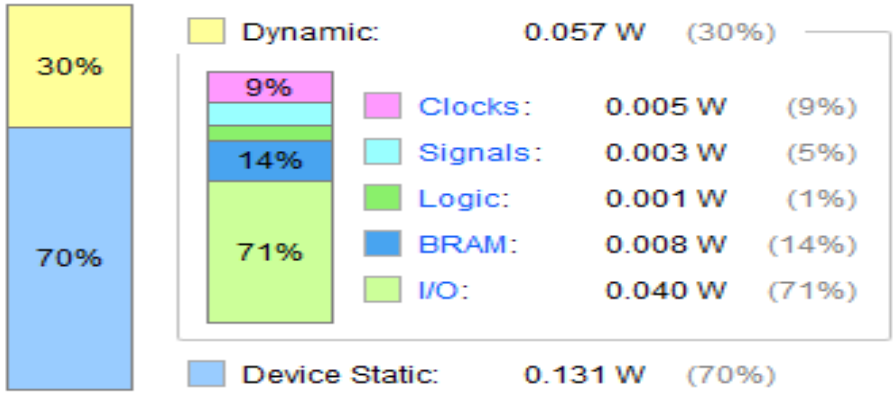




SAIF relativo ad architettura V3 con 4 moduli paralleli (non OOC)
TB con input rampa ordinata di 256 valori da -4 a 4

Total On-Chip Power:	**0.188 W**
Design Power Budget:	**Not Specified**
Process:	typical
Power Budget Margin:	**N/A**
Junction Temperature:	**25,5°C**
Thermal Margin:	59,5°C (22,7 W)
Ambient Temperature:	25.0 °C
Effective ϑJA:	2,6°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	High

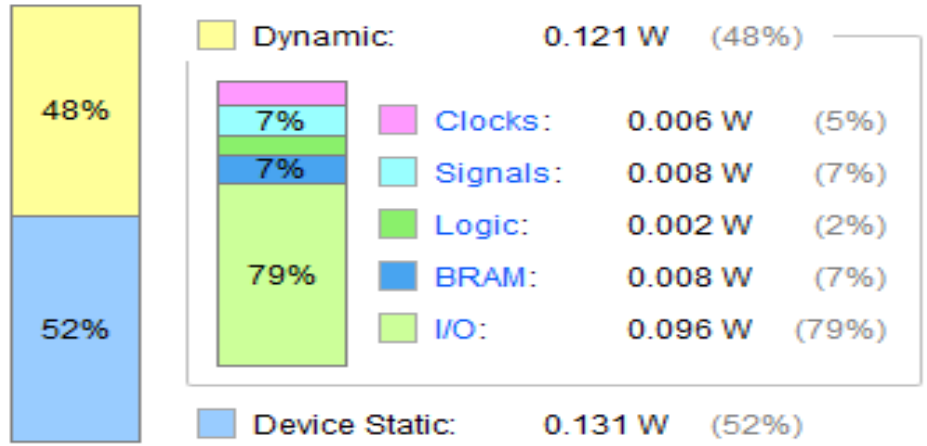
On-Chip Power

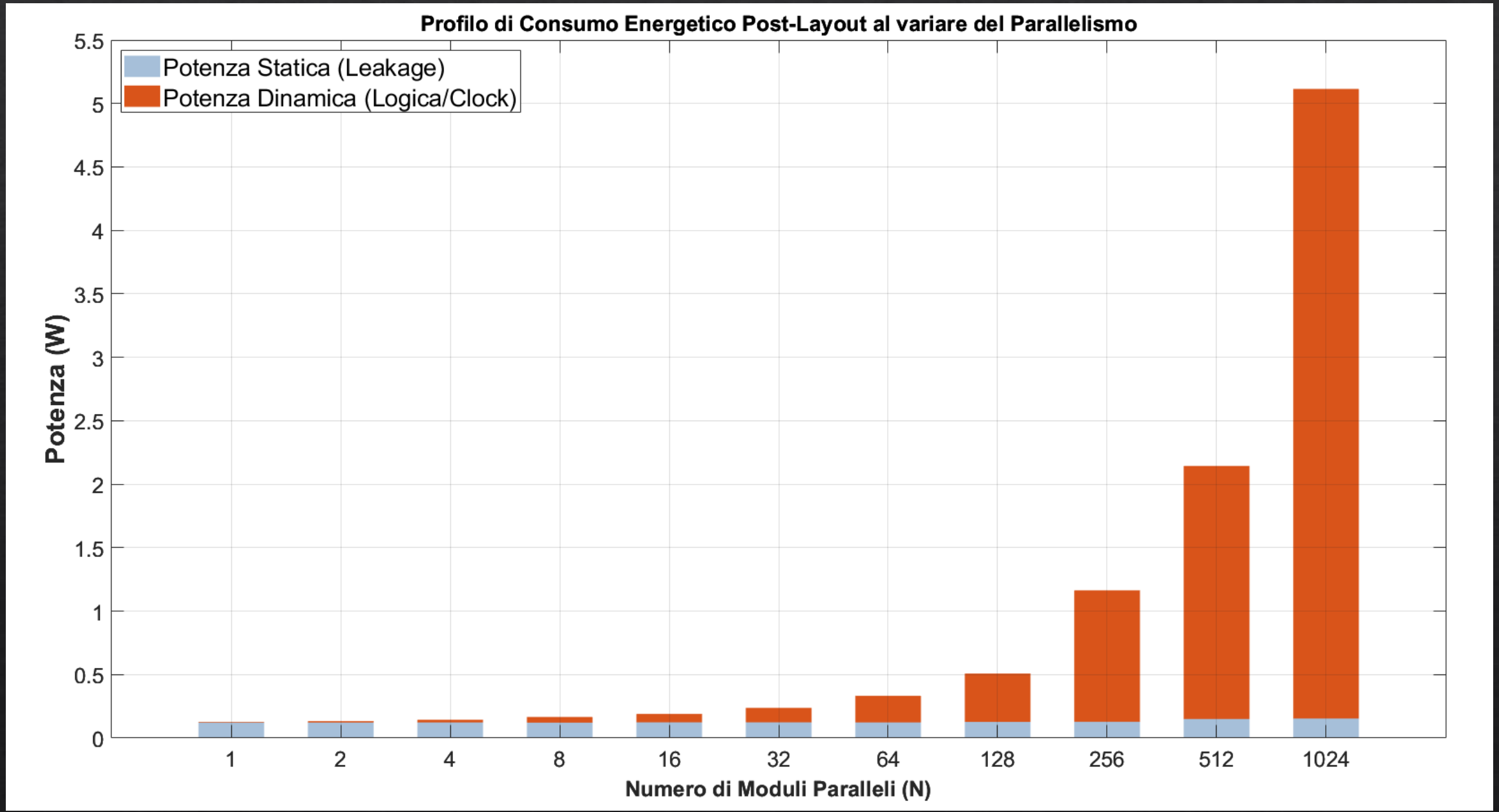


SAIF relativo ad architettura V3 con 4 moduli paralleli (non OOC)
TB con Input reali tensori CIFAR II

Total On-Chip Power:	**0.252 W**
Design Power Budget:	**Not Specified**
Process:	typical
Power Budget Margin:	**N/A**
Junction Temperature:	**25,7°C**
Thermal Margin:	59,3°C (22,6 W)
Ambient Temperature:	25.0 °C
Effective ϑJA:	2,6°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	High

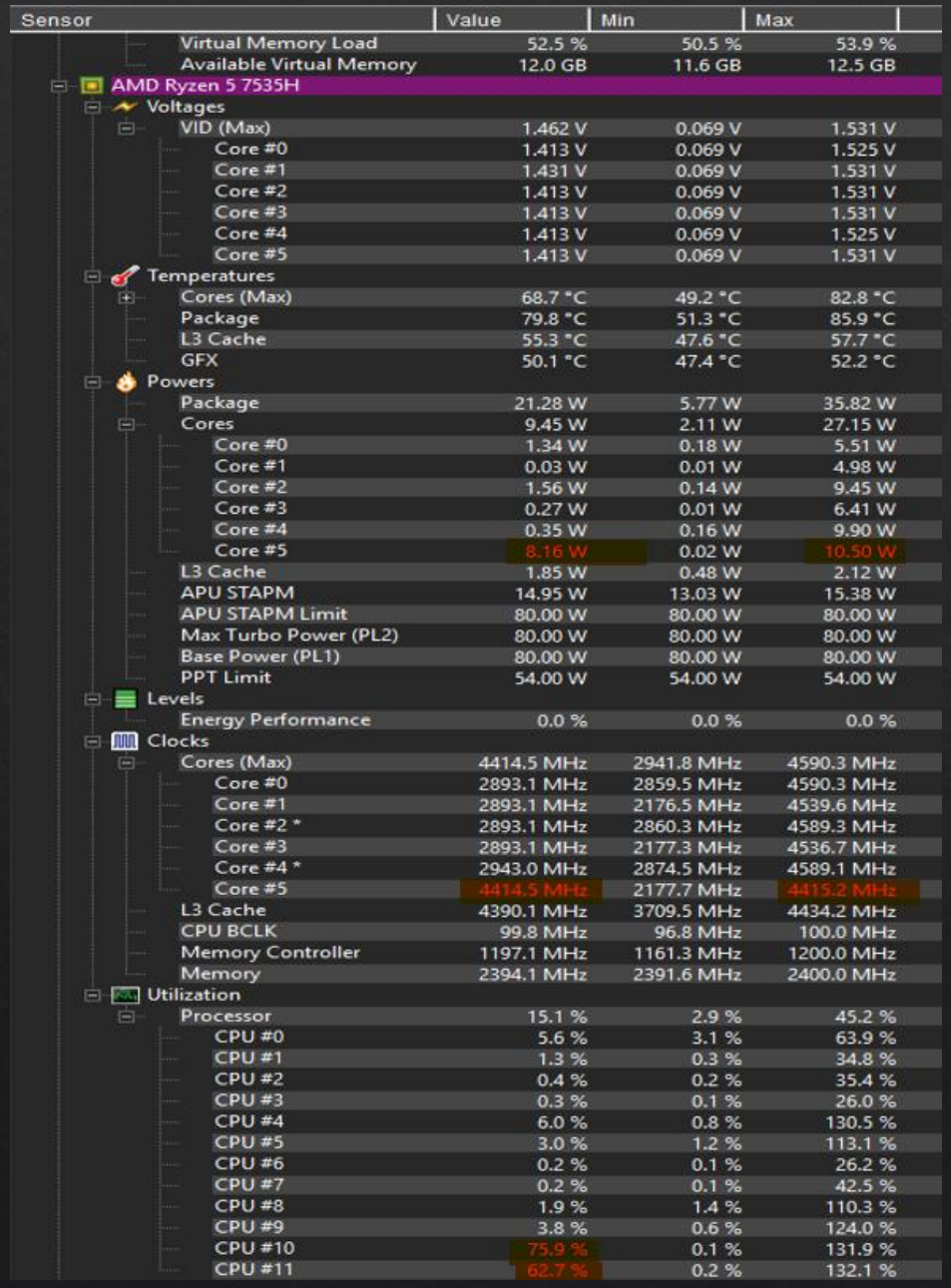
On-Chip Power



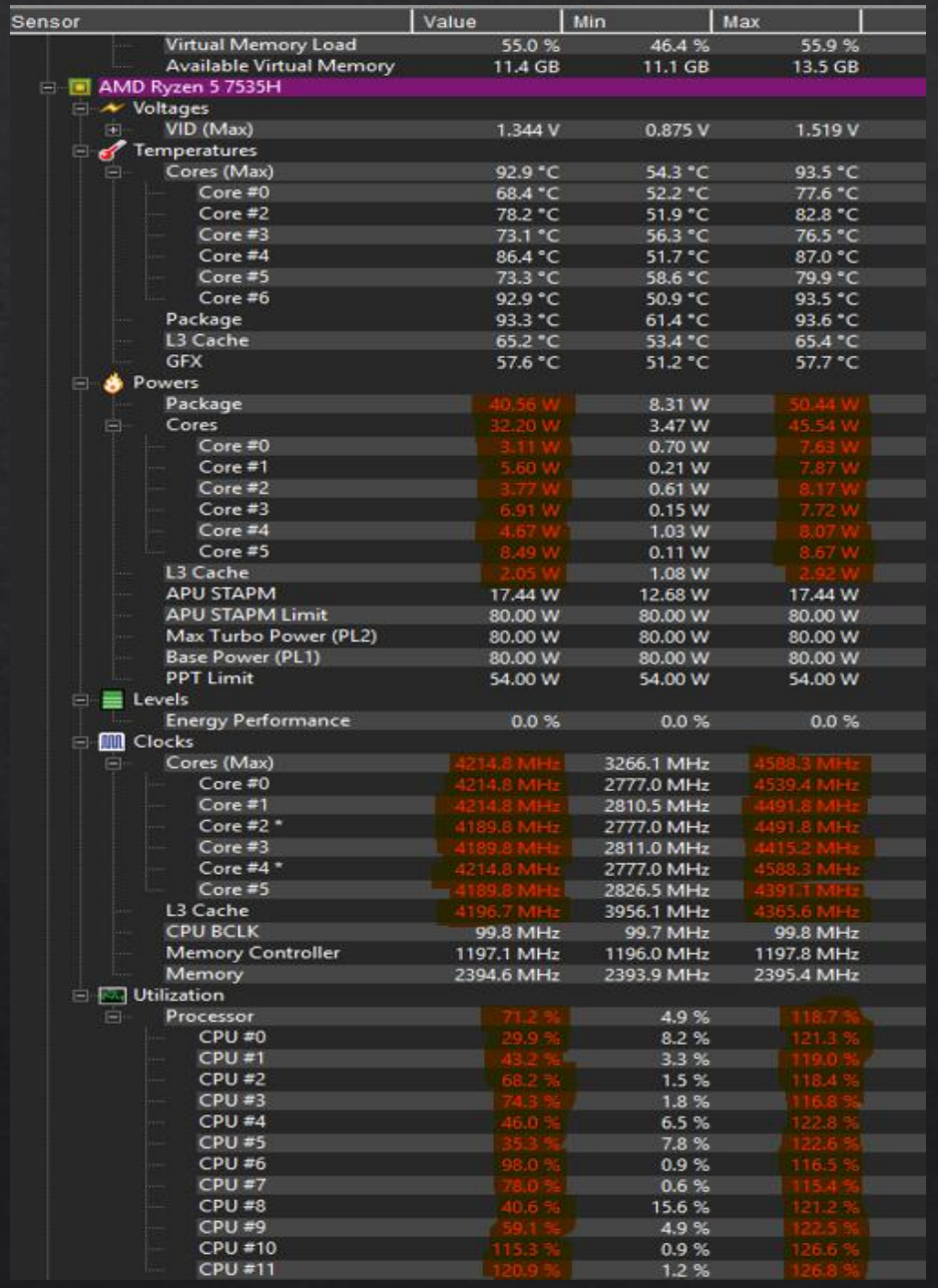

Profilo di Consumo Energetico Post-Layout al variare del Parallelismo
Potenza Statica (Leakage)
Potenza Dinamica (Logica/Clock)
Potenza (W)
5.5
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
1
2
4
8
16
32
64
128
256
512
1024
Numero di Moduli Paralleli (N)

Dettaglio

FPGA vs CPU



Sensor	Value	Min	Max
Virtual Memory Load	52.5 %	50.5 %	53.9 %
Available Virtual Memory	12.0 GB	11.6 GB	12.5 GB
AMD Ryzen 5 7535H			
Voltages			
VID (Max)	1.462 V	0.069 V	1.531 V
Core #0	1.413 V	0.069 V	1.525 V
Core #1	1.431 V	0.069 V	1.531 V
Core #2	1.413 V	0.069 V	1.531 V
Core #3	1.413 V	0.069 V	1.531 V
Core #4	1.413 V	0.069 V	1.525 V
Core #5	1.413 V	0.069 V	1.531 V
Temperatures			
Cores (Max)	68.7 °C	49.2 °C	82.8 °C
Package	79.8 °C	51.3 °C	85.9 °C
L3 Cache	55.3 °C	47.6 °C	57.7 °C
GFX	50.1 °C	47.4 °C	52.2 °C
Powers			
Package	21.28 W	5.77 W	35.82 W
Cores	9.45 W	2.11 W	27.15 W
Core #0	1.34 W	0.18 W	5.51 W
Core #1	0.03 W	0.01 W	4.98 W
Core #2	1.56 W	0.14 W	9.45 W
Core #3	0.27 W	0.01 W	6.41 W
Core #4	0.35 W	0.16 W	9.90 W
Core #5	8.16 W	0.02 W	10.50 W
L3 Cache	1.85 W	0.48 W	2.12 W
APU STAPM	14.95 W	13.03 W	15.38 W
APU STAPM Limit	80.00 W	80.00 W	80.00 W
Max Turbo Power (PL2)	80.00 W	80.00 W	80.00 W
Base Power (PL1)	80.00 W	80.00 W	80.00 W
PPT Limit	54.00 W	54.00 W	54.00 W
Levels			
Energy Performance	0.0 %	0.0 %	0.0 %
Clocks			
Cores (Max)	4414.5 MHz	2941.8 MHz	4590.3 MHz
Core #0	2893.1 MHz	2859.5 MHz	4590.3 MHz
Core #1	2893.1 MHz	2176.5 MHz	4539.6 MHz
Core #2 *	2893.1 MHz	2860.3 MHz	4589.3 MHz
Core #3	2893.1 MHz	2177.3 MHz	4536.7 MHz
Core #4 *	2943.0 MHz	2874.5 MHz	4589.1 MHz
Core #5	4414.5 MHz	2177.7 MHz	4415.2 MHz
L3 Cache	4390.1 MHz	3709.5 MHz	4434.2 MHz
CPU BCLK	99.8 MHz	96.8 MHz	100.0 MHz
Memory Controller	1197.1 MHz	1161.3 MHz	1200.0 MHz
Memory	2394.1 MHz	2391.6 MHz	2400.0 MHz
Utilization			
Processor	15.1 %	2.9 %	45.2 %
CPU #0	5.6 %	3.1 %	63.9 %
CPU #1	1.3 %	0.3 %	34.8 %
CPU #2	0.4 %	0.2 %	35.4 %
CPU #3	0.3 %	0.1 %	26.0 %
CPU #4	6.0 %	0.8 %	130.5 %
CPU #5	3.0 %	1.2 %	113.1 %
CPU #6	0.2 %	0.1 %	26.2 %
CPU #7	0.2 %	0.1 %	42.5 %
CPU #8	1.9 %	1.4 %	110.3 %
CPU #9	3.8 %	0.6 %	124.0 %
CPU #10	75.9 %	0.1 %	131.9 %
CPU #11	62.7 %	0.2 %	132.1 %



Sensor	Value	Min	Max
Virtual Memory Load	55.0 %	46.4 %	55.9 %
Available Virtual Memory	11.4 GB	11.1 GB	13.5 GB
AMD Ryzen 5 7535H			
Voltages			
VID (Max)	1.344 V	0.875 V	1.519 V
Temperatures			
Cores (Max)	92.9 °C	54.3 °C	93.5 °C
Core #0	68.4 °C	52.2 °C	77.6 °C
Core #2	78.2 °C	51.9 °C	82.8 °C
Core #3	73.1 °C	56.3 °C	76.5 °C
Core #4	86.4 °C	51.7 °C	87.0 °C
Core #5	73.3 °C	58.6 °C	79.9 °C
Core #6	92.9 °C	50.9 °C	93.5 °C
Package	93.3 °C	61.4 °C	93.6 °C
L3 Cache	65.2 °C	53.4 °C	65.4 °C
GFX	57.6 °C	51.2 °C	57.7 °C
Powers			
Package	40.56 W	8.31 W	50.44 W
Cores	32.20 W	3.47 W	45.54 W
Core #0	3.11 W	0.70 W	7.63 W
Core #1	5.60 W	0.21 W	7.87 W
Core #2	3.77 W	0.61 W	8.17 W
Core #3	6.91 W	0.15 W	7.72 W
Core #4	4.67 W	1.03 W	8.07 W
Core #5	8.49 W	0.11 W	8.67 W
L3 Cache	2.05 W	1.08 W	2.92 W
APU STAPM	17.44 W	12.68 W	17.44 W
APU STAPM Limit	80.00 W	80.00 W	80.00 W
Max Turbo Power (PL2)	80.00 W	80.00 W	80.00 W
Base Power (PL1)	80.00 W	80.00 W	80.00 W
PPT Limit	54.00 W	54.00 W	54.00 W
Levels			
Energy Performance	0.0 %	0.0 %	0.0 %
Clocks			
Cores (Max)	4214.8 MHz	3266.1 MHz	4588.3 MHz
Core #0	4214.8 MHz	2777.0 MHz	4539.4 MHz
Core #1	4214.8 MHz	2810.5 MHz	4491.8 MHz
Core #2 *	4189.8 MHz	2777.0 MHz	4491.8 MHz
Core #3	4189.8 MHz	2811.0 MHz	4415.2 MHz
Core #4 *	4214.8 MHz	2777.0 MHz	4588.3 MHz
Core #5	4189.8 MHz	2826.5 MHz	4391.1 MHz
L3 Cache	4196.7 MHz	3956.1 MHz	4365.6 MHz
CPU BCLK	99.8 MHz	99.7 MHz	99.8 MHz
Memory Controller	1197.1 MHz	1196.0 MHz	1197.8 MHz
Memory	2394.6 MHz	2393.9 MHz	2395.4 MHz
Utilization			
Processor	71.2 %	4.9 %	118.7 %
CPU #0	29.9 %	8.2 %	121.3 %
CPU #1	43.2 %	3.3 %	119.0 %
CPU #2	68.2 %	1.5 %	118.4 %
CPU #3	74.3 %	1.8 %	116.8 %
CPU #4	46.0 %	6.5 %	122.8 %
CPU #5	35.3 %	7.8 %	122.6 %
CPU #6	98.0 %	0.9 %	116.5 %
CPU #7	78.0 %	0.6 %	115.4 %
CPU #8	40.6 %	15.6 %	121.2 %
CPU #9	59.1 %	4.9 %	122.5 %
CPU #10	115.3 %	0.9 %	126.6 %
CPU #11	120.9 %	1.2 %	126.8 %